PROBLEM4:

PROBLEM STATEMENT:

Employee Payroll and Deduction Analysis: Calculate gross salary from basic pay, allowances, overtime, and incentives; apply progressive deductions and generate a formatted salary statement

OBJECTIVE:

To develop a Python program that calculates an employee's gross salary using basic pay, allowances, overtime, and incentives. The program applies progressive deductions, calculates net salary, and generates a formatted salary statement.

PROBLEM:

```
# Employee Payroll and Deduction Analysisname = input("Enter employee name: ")basic = float(input("Enter basic pay: "))allowances = float(input("Enter allowances: "))overtime = float(input("Enter overtime pay: "))incentives = float(input("Enter incentives: "))if basic < 0 or allowances < 0 or overtime < 0 or incentives < 0:    print("Invalid input: Salary values cannot be negative.")else:     gross = basic + allowances + overtime + incentives # Progressive deductions      if gross <= 20000:       deduction = gross * 0.05        elif gross <= 50000:          deduction = 20000 * 0.05 + (gross - 20000) * 0.10    else:    deduction = (20000 * 0.05 + 30000 * 0.10                    + (gross - 50000) * 0.15)       net_salary = gross - deduction          print("\n========== SALARY STATEMENT ==========")    print("Employee Name :", name)        print("Basic Pay         : Rs.", round(basic, 2))          print("Allowances        : Rs.", round(allowances, 2))    print("Overtime Pay  : Rs.", round(overtime, 2))  print("Incentives        : Rs.", round(incentives, 2))    print("-------------------------------------")  print("Gross Salary  : Rs.", round(gross, 2))  print("Deductions        : Rs.", round(deduction, 2))       print("Net Salary        : Rs.", round(net_salary, 2))  print("=====================================")
```

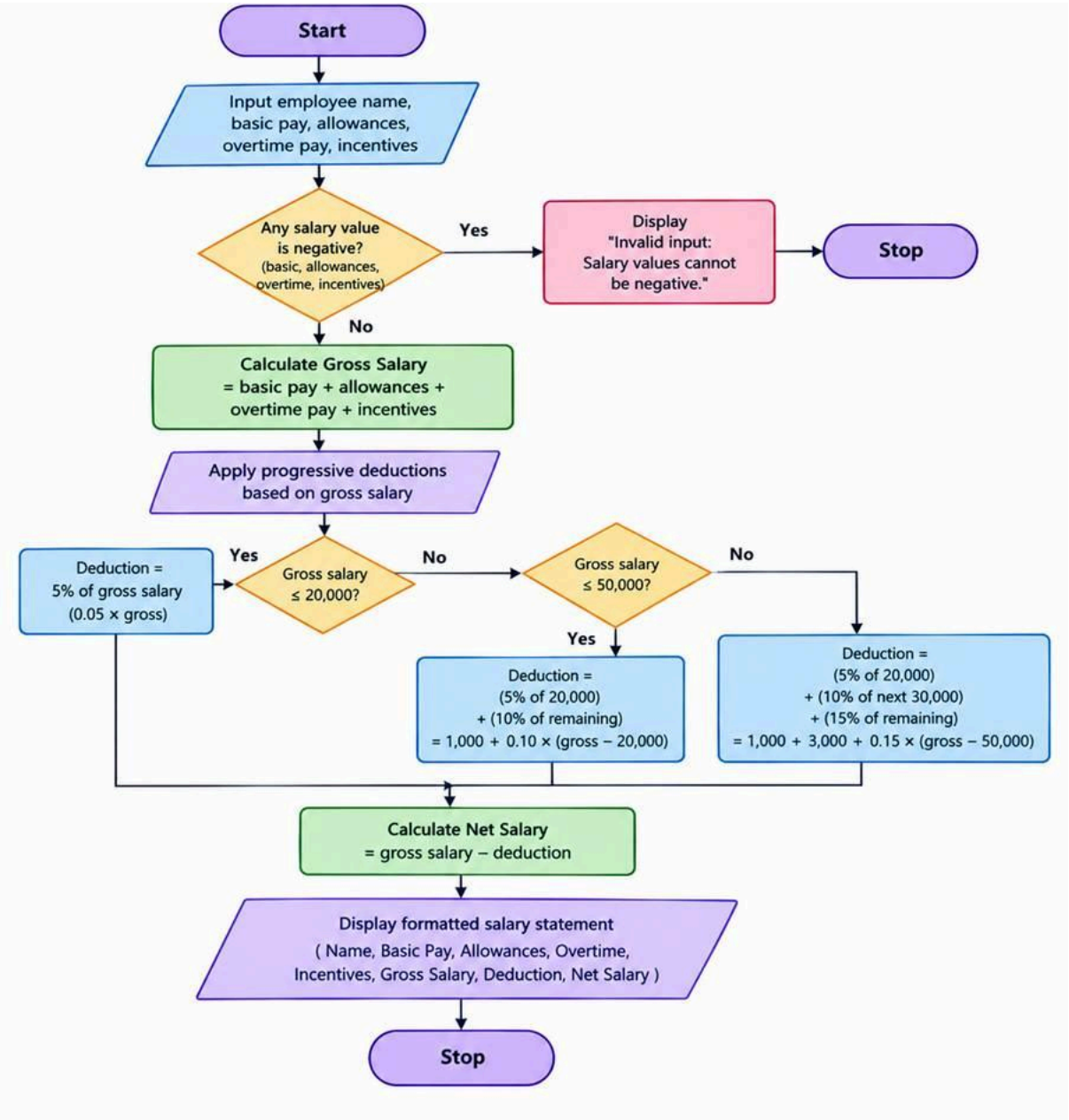
ALGORITHM:

1. Start.
2. Input employee name, basic pay, allowances, overtime pay, and incentives.
3. Validate that all salary values are non-negative.
4. Calculate gross salary by adding basic pay, allowances, overtime, and incentives.
5. Calculate progressive deductions:
    - Up to ₹20,000: 5%.
    - ₹20,001 to ₹50,000: 5% on the first ₹20,000 and 10% on the next ₹30,000.

- Above ₹50,000: apply the previous deductions and 15% on the amount exceeding ₹50,000.

6. Calculate net salary by subtracting deductions from gross salary.
7. Display the formatted salary statement.
8. Stop.

Flowchart



PSEUDOCODE:

BEGIN

INPUT employee_name

INPUT basic_pay, allowances, overtime, incentives

```
        IF any salary value < 0 THEN
        DISPLAY "Invalid input"
        ELSE
        gross = basic_pay + allowances
                + overtime + incentives

        IF gross <= 20000 THEN
        deduction = gross * 5 / 100
        ELSE IF gross <= 50000 THEN
        deduction = 20000 * 5 / 100
                + (gross - 20000) * 10 / 100
        ELSE
        deduction = 20000 * 5 / 100
                + 30000 * 10 / 100
                + (gross - 50000) * 15 / 100
        END IF

        net_salary = gross - deduction

        DISPLAY employee_name
        DISPLAY gross, deduction, net_salary
        END IF
END
```

OUTPUT:

```
Enter employee name: Ravi
Enter basic pay: 30000
Enter allowances: 5000
Enter overtime pay: 2000
Enter incentives: 3000
```

Employee Name : Ravi

Basic Pay : Rs. 30000.0

Allowances : Rs. 5000.0

Overtime Pay : Rs. 2000.0

Incentives : Rs. 3000.0

Gross Salary : Rs. 40000.0

Deductions : Rs. 3000.0

Net Salary : Rs. 37000.0